

Highlights

PhysRoute-Fi: Physically-Routed Two-Branch WiFi-CSI 3D Human Pose Estimation

Jiha Kim,Seunghyun Park,Hyunhee Park

- PhysRoute-Fi: a physically-routed two-branch CSI 3D human pose estimator
- Sanitized phase routes to location; amplitude-led composite branch to posture
- Routing set by measured recoverability, validated component by component
- Enrollment recovers posture-motion amplitude, not frame-accurate tracking
- Selection-clean early-stopping and camera-consistent 2D protocols

ARTICLE INFO

Keywords:

WiFi sensing
channel state information
3D human pose estimation
physical routing
enrollment
motion consistency

ABSTRACT

WiFi channel state information (CSI) supports device-free 3D pose estimation, but posture and motion remain difficult to recover across subjects and scenes. On MM-Fi, recent estimators' Procrustes-aligned errors (99.8–106.2 mm) are close to a *mean-posture reference* (99.4–103.6 mm), indicating that much of the reported gain comes from localization. **PhysRoute-Fi** separates the CSI into physical components and assigns them to pose factors: sanitized per-antenna phase is fused with an amplitude-led composite representation for location, while the composite representation drives a graph-convolutional posture decoder. With one-time enrollment, a motion-consistency loss recovers about 93% of the aggregate inter-frame displacement magnitude and reaches **102.7 mm MPJPE and 65.5 mm PA-MPJPE**; frame-level alignment remains weak. The model is also competitive on scene-benchmark splits (S1-P3: 176.1 vs. 178.5 mm). Evaluation uses a scene-held-out early-stopping rule and a camera-consistent projected-2D metric.

1. Introduction

WiFi CSI is widely used for device-free sensing because it is available from commodity access points and does not require a camera at inference [1, 2]. On MM-Fi [3], CSI-only estimators including MetaFi++ [4], HPE-Li [5], DT-Pose [6], GraphPose-Fi [7], and C-MambaPose [8] have reduced the reported mean per-joint position error (MPJPE).

Posture recovery shows a different pattern. Across the nine official cells (S1/S2/S3 × P1/P2/P3), the PA-MPJPE of methods derived from the official baseline lies in a narrow 99.8–106.2 mm range. A constant training-set mean posture gives 99.4–103.6 mm, whereas the official MetaFi++ baseline gives 116.0–125.4 mm. These values are consistent with predictions that translate a near-average posture through the room. Thus, for the MM-Fi methods examined here, lower absolute MPJPE primarily reflects better localization; articulation and posture motion remain poorly resolved.

PhysRoute-Fi estimates location and root-relative posture in separate paths. A sanitized per-antenna phase branch is fused with an amplitude-led composite branch for location, and a graph decoder maps the composite features to posture. Bone-length and temporal-smoothness terms regularize the two outputs. During one-time enrollment, a displacement loss compares predicted and ground-truth posture changes. Its purpose is to restore motion amplitude, not to assert frame-accurate tracking.

The main contributions are: (i) a two-branch regressor whose routing follows measured component recoverability; (ii) swapped-route, zero-phase, and capacity-matched controls that attribute an 18 mm location gain to two-branch fusion (about 2 mm to sanitized-phase information itself) and a 10 mm posture gain to the graph decoder; and (iii) an enrollment objective that recovers about 93% of aggregate

posture-motion amplitude, although temporal variance is only partly recovered and frame-level alignment is weak. The evaluation also reports a motion-recovery rate, fixes early stopping on a scene-held-out proxy, and uses camera-consistent projected 2D.

Related work. Dedicated RF systems demonstrated through-wall 2D and 3D skeleton recovery [9, 10]. Early commodity-WiFi studies examined pose feasibility and fine-grained person perception [11, 12]; later systems include WiPose [13], WiNect [14], GoPose [15], PerUnet [16], and MetaFi [17]. Person-in-WiFi 3D extends the task to multiple people [18]. On MM-Fi [3], MetaFi++ [4] provides the official CNN–transformer baseline, while HPE-Li [5], DT-Pose [6], GraphPose-Fi [7], and C-MambaPose [8] study lightweight, pretrained, graph-based, and complex-valued models.

Generalization and temporal modeling address related but distinct problems. AdaPose [19] and PerceptAlign [20] target cross-site or cross-layout transfer. RePos [21] separates root-relative pose from root localization, so output factorization itself is not claimed as new here. WiNect [14], MDPose [22], VST-Pose [23], and WiFlow [24] model tracking, velocity, or continuous motion. Video pose and motion-prediction work also shows the value of temporal models and of testing simple static baselines [25, 26]. The present motion claim is narrower: the MM-Fi comparison adds constant-posture anchors and per-sequence alignment diagnostics, and enrollment is evaluated for aggregate motion amplitude rather than precise tracking.

The routing is motivated by prior CSI measurements. Amplitude supports multipath fingerprinting [27]; calibrated phase and spatial differences support localization and tracking [28, 29, 30]; and body-coordinate velocity profiles improve cross-domain sensing [31]. Recent 6-GHz measurements also find amplitude and phase complementary for pose estimation [32]. Wi-Mose [33] is closest to our physical interpretation, associating amplitude with pose and phase with position. PhysRoute-Fi assigns these inputs to explicit

*Corresponding author

■ yaki5896@mju.ac.kr (J. Kim); sp@hansung.ac.kr (S. Park);
hhpark@mju.ac.kr (H. Park)

ORCID(s): 0000-0002-6711-6812 (J. Kim); 0000-0001-5260-1252 (S. Park); 0000-0003-3810-7367 (H. Park)

factor heads and tests that assignment. The enrollment analogy follows fixed-dimensional speaker embeddings such as x-vectors [34].

2. PhysRoute-Fi

2.1. Factorized pose and physical routing

We decompose the pose as

$$Y_t = \mathbf{1}_J r_t^\top + P_t \quad (J=17), \quad (1)$$

where $r_t \in \mathbb{R}^3$ is the location (pelvis) and $P_t \in \mathbb{R}^{17 \times 3}$ is the pelvis-relative posture, and each factor is produced by the physical component that carries it. The routing — sanitized per-antenna phase $\rightarrow$ location, amplitude-led composite branch $\rightarrow$ posture — is set by measured recoverability rather than architectural preference (validated in Section 4.3). Where a component ceases to be recoverable, as the sanitized phase does across unseen scenes, the corresponding path falls back to a statistical anchor rather than a learned regressor.

2.2. Two-branch architecture

Fig. 1 shows the pipeline. The CSI window (300 packets $\times$ 114 subcarriers $\times$ 3 antennas over 30 frames) is split into a **15-channel composite image** — static amplitude (window-mean multipath fingerprint [27]), dynamic amplitude, temporal amplitude-difference, and antenna-pair phase-difference $\Delta\varphi^{(ab)}$ embedded as $\cos/\sin$ (the per-packet offset cancels in the pairwise difference) — and a **6-channel sanitized per-antenna phase image** (each subcarrier phase unwrapped, a least-squares linear trend removed to suppress sampling- and carrier-frequency offsets [28, 29], residual embedded as $\cos/\sin$). Because the composite image also carries antenna-pair phase differences, we call it *composite* rather than amplitude. Each is encoded by a five-block Conv-BN-ReLU branch (stride 2, widths 32/64/128/256/384) with global average pooling into $z_t^a, z_t^p \in \mathbb{R}^{384}$. For the **location**, the two features are fused by a two-token cross-attention [35], $z_t^{\text{loc}} = \text{CrossAttn}(z_t^a, z_t^p)$, and a linear head g_r regresses r_t . For the **posture**, a two-layer graph-convolutional decoder g_p [36] over the 17-joint skeleton maps z_t^a to P_t . The pose is recombined as $\hat{Y}_t = \mathbf{1}_J \hat{r}_t^\top + \hat{P}_t$. The full model has 4.02M parameters and 0.76 GFLOPs per window; training uses AdamW (10^{-3} , batch 256, 40 epochs, cosine annealing, bfloat16).

2.3. Enrollment objective

Each MM-Fi recording (297 frames) is divided into an **enrollment segment** [0, 208) (ground truth captured once), a **44-frame guard gap** that prevents temporal leakage, and an **inference segment** [252, 297) evaluated from CSI alone. All enrollment segments jointly train a single global model — there is no per-recording training or fine-tuning — and all standardization statistics are computed on enrollment frames only. Using the Huber loss ℓ in standardized space with

Table 1

Head configuration per deployment split. In cross-domain modes a factor that is no longer recoverable falls back to a statistical anchor rather than a learned regressor.

Mode	Location	Posture
Enrollment	two-branch fusion	graph decoder
B-S1 (seen)	scene-mean anchor	subject-mean anchor
B-S2 (subj. unseen)	regressed	mean-posture anchor
B-S3 (scene unseen)	no-training anchor	mean-posture anchor

additional forward passes at $f+\delta$ and $f+2\delta$ ($\delta=3$),

$$\begin{aligned} \mathcal{L} = & \ell(g_r(z_f^{\text{loc}}), \bar{r}_f) + \ell(g_p(z_f^a), \bar{P}_f) \\ & + \lambda \ell(g_p(z_{f+\delta}^a) - g_p(z_f^a), \bar{P}_{f+\delta} - \bar{P}_f) \\ & + \beta \mathcal{L}_{\text{bone}} + \gamma \mathcal{L}_{\text{smooth}}, \\ & (\lambda, \beta, \gamma) = (2, 0.5, 0.1), \end{aligned} \quad (2)$$

where $\mathcal{L}_{\text{bone}}$ matches predicted to ground-truth bone lengths over the 16-edge skeleton and $\mathcal{L}_{\text{smooth}}$ penalizes the second temporal difference (an acceleration penalty) in standardized space (full forms in Supplementary S2). The motion-consistency term counters the systematic motion underestimation of mean-seeking estimators: an idealized conditional mean satisfies $\mathbb{E}\|\widehat{\Delta P}\|^2 \leq \mathbb{E}\|\Delta P\|^2$, so the estimated displacement is bounded by the true value and the deficit equals the unexplained motion variance; empirically our regressor exhibits the same deficit (67% motion recovery at $\lambda=0$). We therefore report the **motion-recovery rate**

$$\rho = \mathbb{E}\|\hat{P}_{t+1} - \hat{P}_t\| / \mathbb{E}\|P_{t+1} - P_t\|, \quad (3)$$

the ratio of *aggregate inter-frame displacement magnitude* (posture-motion amplitude, not frame-accurate tracking) alongside MPJPE, without which a near-static prediction can appear preferable to a genuinely moving one.

Under the official cross-domain splits each head instead falls back to what the signal makes recoverable (Table 1). Scene/subject classifiers on z^a reach 99.4% accuracy, so in B-S1 (subjects and scenes seen) both factors use posterior-weighted anchors; in B-S2 (subjects unseen) the location regresses and the posture uses a mean-posture anchor; in B-S3 (scenes unseen) the sanitized phase does not transfer, so the phase branch is gated out and localization reports the no-training anchor (Supplementary S5).

3. Experimental Protocol

MM-Fi [3] contains 4 scenes, 10 subjects, 27 actions, and 297 frames per recording; each frame provides $3 \times 114 \times 10$ CSI samples. We reproduce the official S1 (random subject-action), S2 (cross-subject), and S3 (E01–03 $\rightarrow$ E04 cross-scene) splits. Metrics include MPJPE, PA-MPJPE, root-relative MPJPE, torso-normalized PCK, ρ (Eq. 3), and projected-2D error. Enrollment trains on frames [0, 208), leaves a 44-frame gap, and tests on [252, 297). The early-stopping epoch is fixed on a scene-held-out proxy (train

Inside PhysRoute-Fi — One Forward Pass

From a CSI window to an absolute 17-joint 3D pose

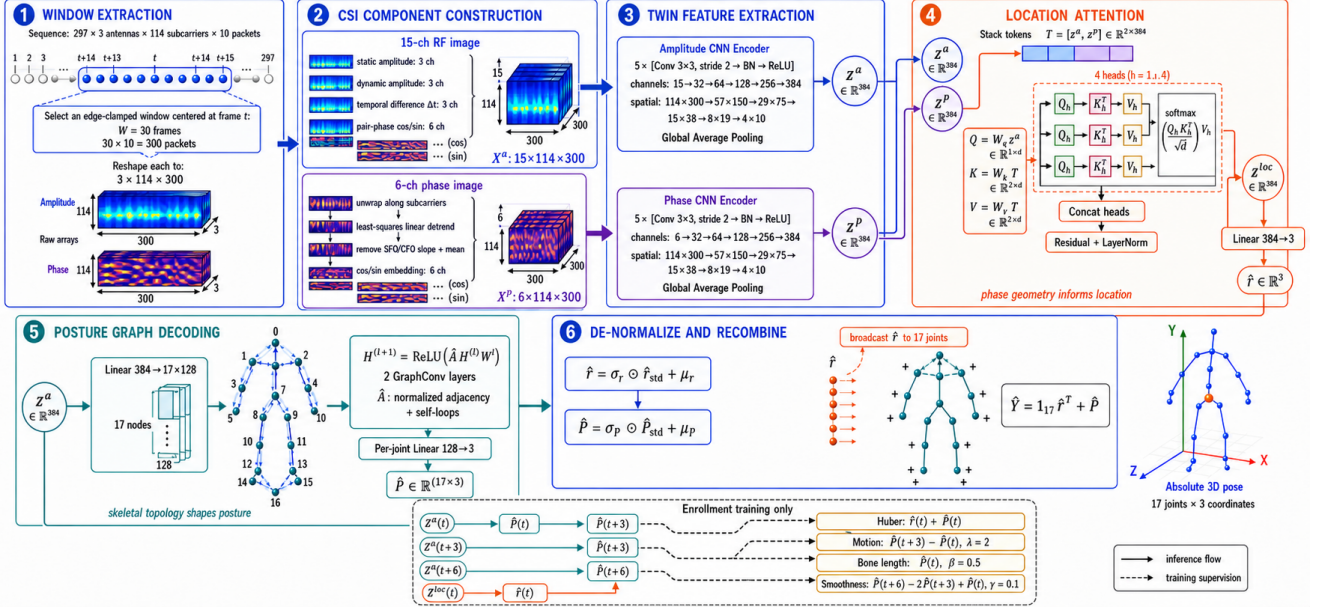


Figure 1: PhysRoute-Fi two-branch routing. The CSI window is split into a 15-channel composite image and a 6-channel sanitized per-antenna phase image, each encoded by a five-block Conv-BN-ReLU branch into $z_t^a, z_t^p \in \mathbb{R}^{384}$. The location head g_r regresses r_t from the cross-attention fusion $z_t^{\text{loc}} = \text{CrossAttn}(z_t^a, z_t^p)$ (composite + sanitized phase); a graph-convolutional posture decoder g_p maps the composite feature z_t^a to P_t ; the pose is recombined as $\hat{Y}_t = \mathbf{1}_{17} \hat{P}_t^T + \hat{P}_t$. Under enrollment, a motion-consistency term aligns predicted frame-to-frame displacement with ground truth. The per-component composite/phase heatmaps are in Supplementary Fig. S1.

E01+E02, validate E03, $e^*=12$) before evaluation; selecting on the evaluation cell can change cross-scene error by 42.3 mm. Training uses AdamW (10^{-3} , weight decay 10^{-4}), batch 256, 40 epochs, cosine annealing, bfloat16, and seed 0. MM-Fi’s 2D and 3D annotations have an ~ 87 px projection residual, so the 2D metric compares predictions with projected ground-truth 3D under a pinhole camera fitted on the training split (Supplementary S6).

4. Results

4.1. Benchmark and anchor diagnosis

On S1-P3, PhysRoute-Fi records **176.1 mm**, compared with 178.5 mm for DT-Pose, and its PA-MPJPE is below the official MM-Fi baseline in all nine cells (full tables in Supplementary S7). At the tighter PCK@10 threshold, it scores 44.2 vs. 33.3 for GraphPose-Fi on S1 and 39.7 vs. 13.1 on S2; the gap is within about two points at coarse thresholds. The anchor rows in Table 2 explain this pattern. A subject-conditioned anchor on S1 and a global anchor on S2 reproduce the PhysRoute-Fi curve within about one point. The precision-band advantage therefore indicates a stable near-mean posture, not framewise articulation. Motion is evaluated only under enrollment (Section 4.2) and only at the amplitude level.

PA-MPJPE remains close to the mean-posture reference for every method, including ours, so cross-domain MPJPE

mainly distinguishes localization. A MetaFi++-style regressor trained in our pipeline retains 21–26% of ground-truth temporal variance (median $\sim 14\%$), has near-zero wrist-trajectory correlation, and gives 105.7–108.3 mm Procrustes error, compared with 101.2–102.7 mm for the constant mean posture. This direct check places the monolithic baseline at or below the constant-posture floor and motivates the separate enrollment analysis.

4.2. Enrollment accuracy and motion amplitude

Enrollment is compared with the mean-posture reference and GraphPose-Fi retrained on the same split (Table 3). PhysRoute-Fi reaches **102.7 mm MPJPE** and **65.5 mm PA-MPJPE** with $\rho=93\%$. Its root-relative and PA errors are 44.3 and 37.6 mm below the reference, respectively; projected-2D error is 10.0 px (PCK2D@50 97.2%). Retrained GraphPose-Fi reaches 104.1 mm MPJPE and 68.0 mm PA-MPJPE but overshoots the displacement amplitude ($\rho=185\%$) at a lower trajectory correlation (0.13 vs. 0.18). The enrollment split accounts for much of the gain below the reference, while the proposed objective controls motion amplitude. Rule-selected good, representative, and difficult sequences are shown in Supplementary Fig. S2.

Figure 2a shows the accuracy–motion trade-off: $\lambda=0$ gives 62.5 mm PA-MPJPE at $\rho=67\%$, $\lambda=4$ gives 68.1 mm at $\rho=97\%$, and $\lambda=2$ is the selected compromise. Because ρ measures magnitude only, we also report displacement-level statistics. At $\lambda=2$, the velocity cosine is 0.09, wrist-trajectory

Table 2

Torso-normalized 3D PCK (%) by threshold (root-relative), protocol P1; upper block S1, lower block S2. A constant near-mean posture (anchor rows, computed from ground truth) reproduces the PhysRoute-Fi curve within about one point, so the precision-band lead over GraphPose-Fi reflects a stable near-mean posture rather than recovered per-frame articulation; the subject-mean anchor is an oracle at S2. On P3, S1-P3 is 176.1 vs. 178.5 mm for DT-Pose (Section 4.1). Best per block in bold, excluding anchor rows.

Method	PCK@10	PCK@20
<i>S1 (seen scenes and subjects)</i>		
GraphPose-Fi [7]	33.3	61.1
PhysRoute-Fi (Benchmark)	44.2	65.9
<i>Mean-posture anchor</i>	<i>41.1</i>	<i>64.4</i>
<i>Subject-mean anchor</i>	<i>44.2</i>	<i>66.2</i>
<i>S2 (cross-subject)</i>		
GraphPose-Fi [7]	13.1	44.2
PhysRoute-Fi (B-S2)	39.7	64.9
<i>Mean-posture anchor</i>	<i>40.6</i>	<i>65.6</i>
<i>Subject-mean anchor (oracle)</i>	<i>48.5</i>	<i>67.8</i>

correlation is 0.18, and the predicted-to-true posture variance ratio is 0.58. Across 810 sequences, wrist correlation has median 0.14 and IQR 0.05–0.27; the illustrated $r=0.85$ sequence is near the 99.9th percentile (Fig. 2b). Thus, the model recovers motion amplitude and part of its variance, but not reliable frame-to-frame velocity.

The enrollment burden is strongly data-dependent (Supplementary S9). With 10 labeled frames per sequence, root-relative error is 117.5 mm, effectively the 117.4 mm mean-posture reference; PA-MPJPE then falls to 92.3, 85.4, 76.4, and 65.5 mm with 30, 60, 120, and 208 frames, respectively. The full segment is about 20 s of captured motion. Guard gaps of 0, 44, and 88 frames give 60.3, 65.5, and 71.5 mm PA-MPJPE; we retain 44 frames to reduce adjacent-segment leakage despite the 5.2 mm penalty. At the smallest budget, $\rho=178\%$ reflects jitter rather than improved tracking.

4.3. Essential routing controls

Table 4 adds each route separately. The phase branch and fusion reduce pelvis error from 84.0 to 65.6 mm without changing root-relative posture; the graph decoder reduces PA-MPJPE from 77.2 to 67.2 mm without changing pelvis error. Table 3 provides three further controls. Swapping the routes raises PA-MPJPE from 65.5 to 70.5 mm. Replacing the GCN with a parameter-matched MLP (870.7k vs. 871.2k parameters) raises it to 68.0 mm. With fusion retained but phase set to zero during training and evaluation, pelvis error is 67.7 mm. Thus, fusion recovers most of the 18.2 mm location gain from the composite input, and sanitized phase contributes about 2 mm in the enrolled scene (67.7 vs. 65.7 mm). Evidence specific to phase comes from this control, its failure to transfer across scenes, and the analyses in Supplementary S4. The GraphPose-Fi and zero-phase rows are single-seed.

Table 3

Enrollment results and essential routing controls (hold-out inference segments, all CSI-only). Upper block: enrollment main comparison — mean-posture reference, retrained GraphPose-Fi, and PhysRoute-Fi at $\lambda=0$ and $\lambda=2$. Lower block: routing controls, where the first numeric column reports the *pelvis/root* error rather than absolute MPJPE (the zero-phase control keeps the fusion stage but feeds an all-zero phase branch). The incremental single-branch→two-branch route validation is in Supplementary S10. $^\dagger \rho>100\%$ reflects frame-to-frame jitter, not tracked motion. The GraphPose-Fi row and the zero-phase control are single-seed. Main-model 3-seed statistics: MPJPE 103.1 ± 0.6 , PA 66.0 ± 0.5 , ρ $93.7 \pm 2.4\%$.

Model	MPJPE	Root-rel.	PA	ρ	Wrist r
Mean-posture reference	—	117.4	103.1	0%	0
GraphPose-Fi (retrained) [7]	104.1	77.0	68.0	185% [†]	0.13
PhysRoute-Fi ($\lambda=0$)	98.4	69.7	62.5	67%	0.17
PhysRoute-Fi ($\lambda=2$, full)	102.7	73.1	65.5	93%	0.18
<i>Routing controls (first column = pelvis/root error, mm)</i>					
PhysRoute-Fi (full)	65.7	73.1	65.5	93%	0.18
zero-phase (fusion, no phase)	67.7	73.1	65.5	90%	—
swapped routing	66.3	78.6	70.5	88%	—
capacity-matched MLP decoder	65.3	75.5	68.0	84%	—

Table 4

Incremental route validation in enrollment mode (mm). Each route improves its assigned factor while leaving the other essentially unchanged.

Configuration	Pelvis	PA
Single-branch baseline	84.0	77.2
+ phase branch/fusion	65.6	≈77.2
+ graph decoder	≈84.0	67.2
PhysRoute-Fi (both)	65.7	65.5

The component ablation in Supplementary S8 identifies static amplitude as the main location input; removing temporal amplitude-difference or antenna-pair phase-difference reduces ρ to 70% or 66%. At test time, zeroing 10/20/30% of subcarriers increases absolute error by 26/56/82 mm, and removing one antenna roughly doubles it. Gaussian phase jitter of $\sigma=0.1/0.3/0.5$ rad raises pelvis error from 65.7 mm to 65.9/70.1/87.0 mm, while changing root-relative and PA errors by less than 0.1 mm. These single-seed stress tests localize the main fragility to the phase-to-location path. Supplementary S4 reports input-gradient saliency [37], attention weights, t-SNE embeddings [38], and a mixture-density uncertainty head [39]; these analyses are consistent with, but do not prove, the proposed routing.

5. Limitations and Conclusion

Enrollment requires ground-truth capture and applies only to enrolled subject–action pairs. When 21 actions are enrolled and six are held out, PA-MPJPE rises to 107.7–113.5 mm, compared with 59.2–80.2 mm for the enrolled folds; the learned posture does not transfer to unseen actions. MM-Fi has one recording per subject–action pair, so enrollment and test segments come from the same session,

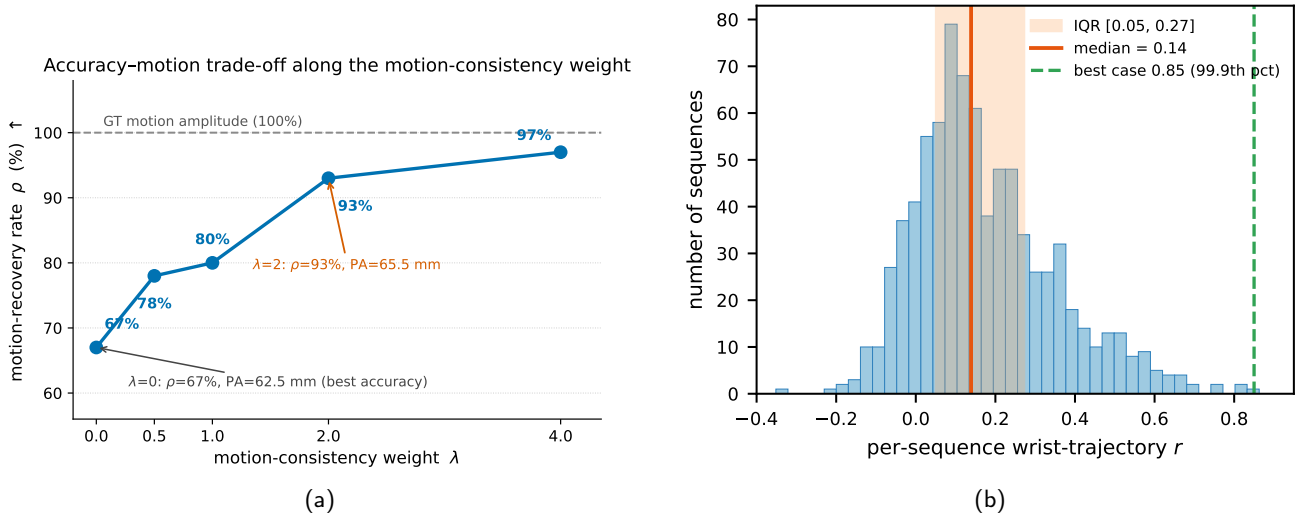


Figure 2: Motion fidelity under enrollment. (a) Motion-recovery rate ρ vs. the motion-consistency weight λ , against the ground-truth amplitude (100%); ρ is an aggregate inter-frame displacement-magnitude ratio, not frame-accurate tracking. (b) Per-sequence right-wrist trajectory correlation r over 810 validation sequences: the distribution is centered at median **0.14** with IQR **0.05–0.27**; the $r=0.85$ best case of the qualitative sequence (Supplementary Fig. S3) sits near the 99.9th percentile, consistent with motion recovered at the level of amplitude rather than frame-accurate tracking.

separated by the guard gap. Independent-session evaluation is not possible with this dataset. The protocol also assumes frame-aligned CSI and pose, and S3 changes both scene and subject. The phase branch does not beat the no-training anchor across scenes. Results are limited to MM-Fi; the main enrollment result varies by at most ± 0.6 mm across three seeds, but ablations use one seed and ρ is epoch-sensitive. The 0.09 velocity cosine and 0.18 wrist correlation further restrict the result to motion amplitude. A stable near-mean pose may serve as a low-variance camera-free prior, but it is not a tracking solution.

PhysRoute-Fi separates location and posture according to measured CSI-component recoverability and tests the assignment with factor-specific controls. On seen-scene MM-Fi splits it remains competitive with larger recent models. Under enrollment it obtains 102.7 mm MPJPE, 65.5 mm PA-MPJPE, and about 93% aggregate motion amplitude. Frame-accurate temporal alignment and cross-scene localization remain unresolved.

Acknowledgment

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (2022R1A2C2005705, AI-MAC Protocol on Distributed Machine Learning for Intelligent Flying Base Station).

CRedit authorship contribution statement

Jiha Kim: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Seunghyun Park:** Validation, Data curation, Writing – review & editing.

Hyunhee Park: Supervision, Project administration, Funding acquisition, Writing – review & editing.

References

- [1] Y. Ma, G. Zhou, S. Wang, WiFi sensing with channel state information: A survey, *ACM Computing Surveys* 52 (2019) 1–36. doi:10.1145/3310194.
- [2] J. Yang, X. Chen, H. Zou, C. X. Lu, D. Wang, S. Sun, L. Xie, SenseFi: A library and benchmark on deep-learning-empowered WiFi human sensing, *Patterns* 4 (2023) 1–22. doi:10.1016/j.patter.2023.100703.
- [3] J. Yang, H. Huang, Y. Zhou, X. Chen, Y. Xu, S. Yuan, H. Zou, C. X. Lu, L. Xie, MM-Fi: Multi-modal non-intrusive 4D human dataset for versatile wireless sensing, in: *Advances in Neural Information Processing Systems*, volume 36, 2023, pp. 18756–18768. arXiv:2305.10345, datasets and Benchmarks Track.
- [4] Y. Zhou, H. Huang, S. Yuan, H. Zou, L. Xie, J. Yang, MetaFi++: WiFi-enabled transformer-based human pose estimation for metaverse avatar simulation, *IEEE Internet of Things Journal* 10 (2023) 14128–14136. doi:10.1109/JIOT.2023.3262940.
- [5] T. D. Gian, T. D. Lai, T. V. Luong, K.-S. Wong, V.-D. Nguyen, HPE-Li: WiFi-enabled lightweight dual selective kernel convolution for human pose estimation, in: *Computer Vision – ECCV 2024*, volume 15089 of *Lecture Notes in Computer Science*, Springer Nature Switzerland, 2025, pp. 93–111. doi:10.1007/978-3-031-72751-1_6, first published online on 26 October 2024; presented at ECCV 2024.
- [6] Y. Chen, J. Guo, DT-Pose: Towards robust and realistic human pose estimation using WiFi signals, *Edge Intelligence and Systems* 1 (2026) 2. Preprint: arXiv:2501.09411.
- [7] J. Chen, Y. Qu, R. Tang, D. Slock, Graph-based 3D human pose estimation using WiFi signals, in: *2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Barcelona, Spain, 2026, pp. 19992–19996. doi:10.1109/ICASSP55912.2026.11463884. arXiv:2511.19105.
- [8] Phuc Nguyen H, C-MambaPose: A physics-informed complex mamba framework for cross-environment WiFi human pose estimation, 2026. doi:10.48550/arXiv.2606.13700. arXiv:2606.13700, arXiv preprint.
- [9] M. Zhao, T. Li, M. Abu Alsheikh, Y. Tian, H. Zhao, A. Torralba, D. Katabi, Through-wall human pose estimation using radio signals,

- in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2018, pp. 7356–7365. doi:10.1109/CVPR.2018.00768.
- [10] M. Zhao, Y. Tian, H. Zhao, M. Abu Alsheikh, T. Li, R. Hristov, Z. Kabelac, D. Katabi, A. Torralba, RF-based 3D skeletons, in: Proceedings of the 2018 Conference of the ACM Special Interest Group on Data Communication (SIGCOMM), Association for Computing Machinery, 2018, pp. 267–281. doi:10.1145/3230543.3230579.
- [11] F. Wang, S. Panev, Z. Dai, J. Han, D. Huang, Can WiFi estimate person pose?, 2019. doi:10.48550/arXiv.1904.00277. arXiv:1904.00277, arXiv preprint.
- [12] F. Wang, S. Zhou, S. Panev, J. Han, D. Huang, Person-in-WiFi: Fine-grained person perception using WiFi, in: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2019, pp. 5452–5461. doi:10.1109/ICCV.2019.00555.
- [13] W. Jiang, H. Xue, C. Miao, S. Wang, S. Lin, C. Tian, S. Murali, H. Hu, Z. Sun, L. Su, Towards 3D human pose construction using WiFi, in: Proceedings of the 26th Annual International Conference on Mobile Computing and Networking (MobiCom), Association for Computing Machinery, 2020, pp. 1–14. doi:10.1145/3372224.3380900.
- [14] Y. Ren, Z. Wang, S. Tan, Y. Chen, J. Yang, WiNect: 3D human pose tracking for free-form activity using commodity WiFi, Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies 5 (2021) 1–29. doi:10.1145/3494973.
- [15] Y. Ren, Z. Wang, Y. Wang, S. Tan, Y. Chen, J. Yang, GoPose: 3D human pose estimation using WiFi, Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies 6 (2022) 1–25. doi:10.1145/3534605.
- [16] Y. Zhou, A. Zhu, C. Xu, F. Hu, Y. Li, PerUnet: Deep signal channel attention in UNet for WiFi-based human pose estimation, IEEE Sensors Journal 22 (2022) 19750–19760. doi:10.1109/JSEN.2022.3204607.
- [17] J. Yang, Y. Zhou, H. Huang, H. Zou, L. Xie, MetaFi: Device-free pose estimation via commodity WiFi for metaverse avatar simulation, in: 2022 IEEE 8th World Forum on Internet of Things (WF-IoT), 2022, pp. 1–6. doi:10.1109/WF-IoT54382.2022.10152057.
- [18] K. Yan, F. Wang, B. Qian, H. Ding, J. Han, X. Wei, Person-in-WiFi 3D: End-to-end multi-person 3D pose estimation with Wi-Fi, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, pp. 969–978. doi:10.1109/CVPR52733.2024.00098.
- [19] Y. Zhou, J. Yang, H. Huang, L. Xie, AdaPose: Toward cross-site device-free human pose estimation with commodity WiFi, IEEE Internet of Things Journal 11 (2024) 40255–40267. doi:10.1109/JIOT.2024.3452670.
- [20] S. Jia, Y. Lu, B. Liu, X. Zhang, P. Zhao, X. Tang, Y. Wei, J. Huang, H. Yan, Z. Liu, Breaking coordinate overfitting: Geometry-aware WiFi sensing for cross-layout 3D pose estimation, 2026. arXiv:2601.12252, arXiv preprint.
- [21] Z. Hou, T. Ohtsuki, RePos: Relative-to-absolute output factorization for cross-environment WiFi-based 3D human pose estimation, 2026. doi:10.48550/arXiv.2607.02986. arXiv:2607.02986, arXiv preprint.
- [22] C. Tang, W. Li, S. Vishwakarma, F. Shi, S. Julier, K. Chetty, MDPose: Human skeletal motion reconstruction using WiFi micro-doppler signatures, IEEE Transactions on Aerospace and Electronic Systems 60 (2024) 157–167. doi:10.1109/TAES.2023.3256973.
- [23] X. Zhang, Z. Ye, J. Zhang, X. Tian, Z. Liang, S. Yu, VST-Pose: A velocity-integrated spatiotemporal attention network for human WiFi pose estimation, 2025. arXiv:2507.09672, arXiv preprint.
- [24] Y. Dao, L. Zhang, H. Liu, H. Zhang, W. Wang, WiFlow: A lightweight WiFi-based continuous human pose estimation network with spatio-temporal feature decoupling, 2026. doi:10.48550/arXiv.2602.08661. arXiv:2602.08661, arXiv preprint.
- [25] D. Pavlo, C. Feichtenhofer, D. Grangier, M. Auli, 3D human pose estimation in video with temporal convolutions and semi-supervised training, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019, pp. 7753–7762. doi:10.1109/CVPR.2019.00794.
- [26] J. Martinez, M. J. Black, J. Romero, On human motion prediction using recurrent neural networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017, pp. 2891–2900. doi:10.1109/CVPR.2017.497.
- [27] X. Wang, L. Gao, S. Mao, S. Pandey, DeepFi: Deep learning for indoor fingerprinting using channel state information, in: 2015 IEEE Wireless Communications and Networking Conference (WCNC), 2015, pp. 1666–1671. doi:10.1109/WCNC.2015.7127718.
- [28] X. Wang, L. Gao, S. Mao, PhaseFi: Phase fingerprinting for indoor localization with a deep learning approach, in: 2015 IEEE Global Communications Conference (GLOBECOM), 2015, pp. 1–6. doi:10.1109/GLOCOM.2015.7417517.
- [29] M. Kotaru, K. Joshi, D. Bharadia, S. Katti, SpotFi: Decimeter level localization using WiFi, in: Proceedings of the 2015 ACM Conference on Special Interest Group on Data Communication (SIGCOMM), Association for Computing Machinery, 2015, pp. 269–282. doi:10.1145/2785956.2787487.
- [30] K. Qian, C. Wu, Z. Yang, Y. Liu, K. Jamieson, Widar: Decimeter-level passive tracking via velocity monitoring with commodity Wi-Fi, in: Proceedings of the 18th ACM International Symposium on Mobile Ad Hoc Networking and Computing (MobiHoc), Association for Computing Machinery, 2017, pp. 1–10. doi:10.1145/3084041.3084067.
- [31] Y. Zhang, Y. Zheng, K. Qian, G. Zhang, Y. Liu, C. Wu, Z. Yang, Widar3.0: Zero-effort cross-domain gesture recognition with Wi-Fi, IEEE Transactions on Pattern Analysis and Machine Intelligence 44 (2022) 8671–8688. doi:10.1109/TPAMI.2021.3105387.
- [32] Z. Yin, M. Brambilla, M. Nicoli, Human walking sensing and pose estimation in the 6 GHz band using amplitude and phase CSI, 2026. arXiv:2606.10048, arXiv preprint.
- [33] Y. Wang, L. Guo, Z. Lu, X. Wen, S. Zhou, W. Meng, From point to space: 3D moving human pose estimation using commodity WiFi, IEEE Communications Letters 25 (2021) 2235–2239. doi:10.1109/LCOMM.2021.3073271.
- [34] D. Snyder, D. Garcia-Romero, G. Sell, D. Povey, S. Khudanpur, X-Vectors: Robust DNN embeddings for speaker recognition, in: 2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2018, pp. 5329–5333. doi:10.1109/ICASSP.2018.8461375.
- [35] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, in: Advances in Neural Information Processing Systems, volume 30, 2017, pp. 5998–6008.
- [36] L. Zhao, X. Peng, Y. Tian, M. Kapadia, D. N. Metaxas, Semantic graph convolutional networks for 3D human pose regression, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019, pp. 3425–3435. doi:10.1109/CVPR.2019.00354.
- [37] K. Simonyan, A. Vedaldi, A. Zisserman, Deep inside convolutional networks: Visualising image classification models and saliency maps, 2014. doi:10.48550/arXiv.1312.6034. arXiv:1312.6034, 2nd International Conference on Learning Representations (ICLR), Workshop Track Proceedings.
- [38] L. van der Maaten, G. Hinton, Visualizing data using t-SNE, Journal of Machine Learning Research 9 (2008) 2579–2605.
- [39] C. M. Bishop, Mixture Density Networks, Technical Report NCRG/94/004, Aston University, 1994.